# LESSON PLAN

## ENGINE HEALTH MONITORING (9)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 01 |
| **Total No. of Lesson Plans** | 7 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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| **1. Topic** | Introductions to various standards BS, AS, DTD, MIL, AN, NAS, close tolerance, unified etc. |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Introductions to various standards BS, AS, DTD, MIL, AN, NAS, close tolerance, unified etc. |
| **5. Specific Objectives** | 9.1.1.1 Trainee will be able to describe the use of various standards, identification of bolts, nuts, safety methods and maintenance of cables. |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Review of standard identification charts; Instructor supervision required for safety method verification. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What does the acronym MIL stand for in engineering standards?
(b) Can you list three common aviation bolt standards?
(c) Why is it critical to identify the correct standard before maintenance?

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • British Standard (BS) specifications<br>• American Society for Testing and Materials (ASTM)<br>• Department of Transportation (DOT) standards<br>• Military Standard (MIL-Spec) overview<br>• National Aerospace Standard (NAS) definitions | L & Ds | PPT, IB | 15 min |
| (b) | • AN (Air Force-Navy) standard bolt features<br>• Unified National Coarse and Fine threads<br>• Close tolerance bolt identification marks<br>• Safety methods for cable maintenance<br>• Standard nut types and locking mechanisms | L & Ds | PPT, PDF | 20 min |
| (c) | • Distinction between DTD and NAS standards<br>• Application of close tolerance in critical joints<br>• Safety protocols for cable tensioning<br>• Maintenance procedures for standard fasteners<br>• Summary of standard selection criteria | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the primary difference between an AN standard and a NAS standard.
(b) Describe one safety method used during cable maintenance to prevent failure.
(c) Identify two characteristics that define a close tolerance bolt.

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| **12. Home Assignment and Follow Up** | Prepare a brief note comparing the usage of MIL standards versus NAS standards for aircraft fasteners, referencing LP 02 for specific examples. |
| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Identification of AN-type, NAS etc. grip length, torques, a/c bolt identification, screws, taper pins. | Refer to the master syllabus reference list | LP 02 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min